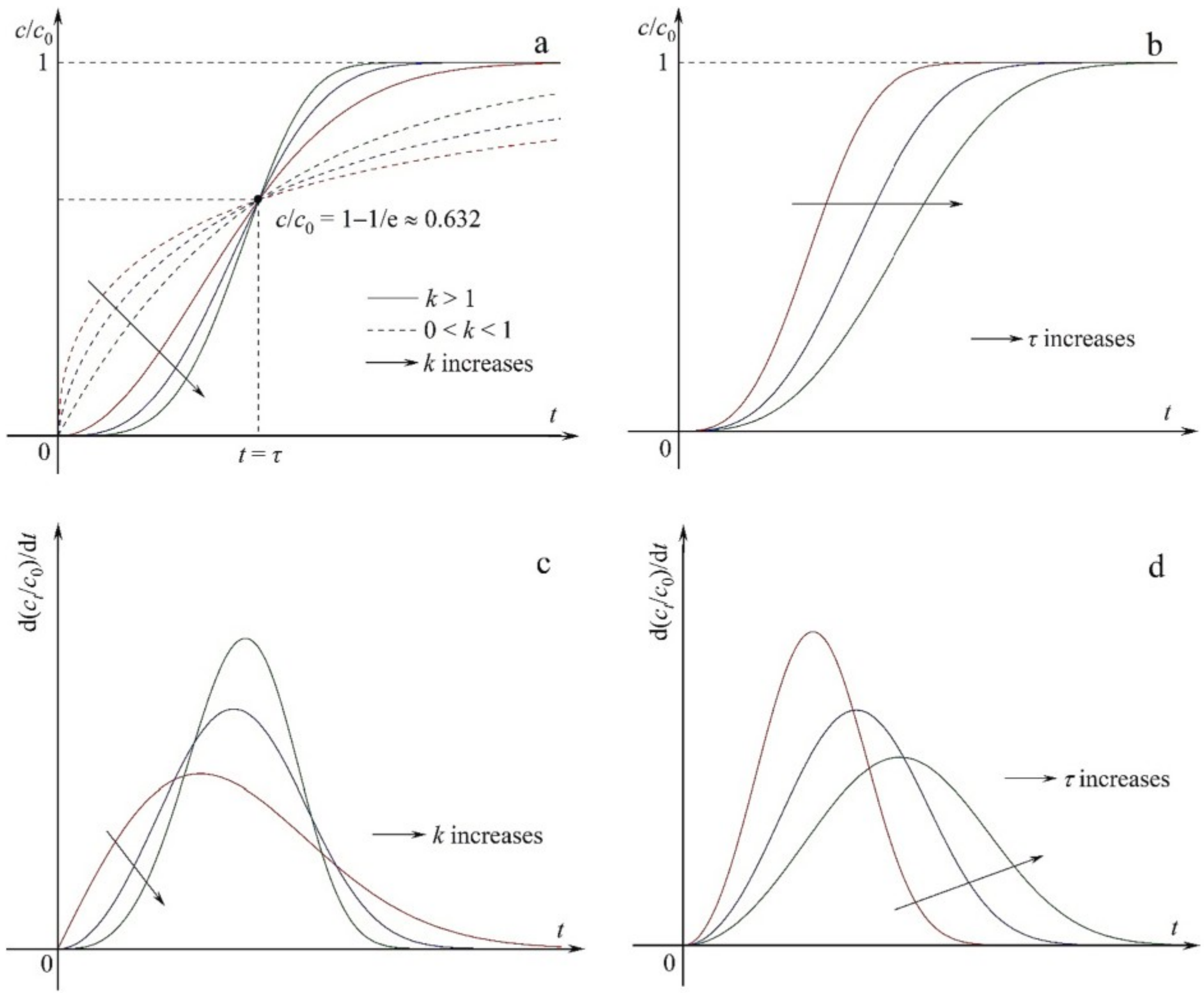

Fig. 4. Curve characteristics of the Weibull model and its first-order derivative.

It should be noted that the logarithmic terms that contain time t in Eqs. (49)–(53) keep dimensionless by introducing a set of variables/parameters with a value of unity artificially. This modification is meaningless in practice. In fact, the reason why Eqs. (49)–(53) have the ability to describe the asymmetric breakthrough curve is because they change the mathematical structure of the original models. To some extent, Eqs. (49)–(51) lose their theoretical basis and the model parameters also lack the physical meanings. From a mathematical perspective, the value of all logarithmic terms must be more than zero, i.e. $t > 1/c_0$ Eq. (49) and Eq. (50) or $t > 1$ for Eqs. (51)–(53). In this case, the fitted curve may deviate from the breakthrough curves in the initial stage of adsorption.

Recently, Singh et al. propose an empirical breakthrough model by a combination of the Avrami equation with the breakthrough curves characteristics, which is expressed as [114].

$$\frac{c}{c_0} = 1 - exp(-kt^n) \tag{54}$$

One of its advantages is that the relative concentration c/c_0 is also equal to zero at $t = 0$. As shown in Fig. S2, the parameters k and n can influence the degree of curvature and position of the curves simultaneously. The value of n determines the type of the function curve, representing an asymmetric S-shaped curve at $n > 1$ and a L-shaped curve at $0 < n \leq 1$. Two adjustable parameters k and n make the curve fitting more flexible. Thus, Eq. (54) may be expected to provide the high fitting accuracy for description of many breakthrough curves.

4. Comprehensive evaluation of fitting quality

Although there are many mechanistic and empirical mathematical models available for modeling of the breakthrough curves in a fixed-bed column, their fitting quality is not sufficiently evaluated. In most fixed-bed studies, the selection of the breakthrough models seems considerably arbitrary when correlated with the experimental data. Recently, Kimani compare the fitting performance of some breakthrough models systematically based on the measured breakthrough data from bisphenol A adsorption on polyaniline [115]. In general, a good fit requires that the distribution of the experimental data should match the curve characteristics of the breakthrough models well [43]. The fitting quality of the breakthrough models can be visually evaluated by how close the fitted curve is to the data points. If the curve fitting succeeds, a plot of the predicted versus observed values will produce a straight line that evenly passes through the data points [104]. In addition, error statistics are also used to quantitatively assess the fitting quality of the breakthrough models. The coefficient of determination (R^2) is a good measure of the goodness of fit, which is expressed as [25].

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \acute{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y}_i)^2} \tag{55}$$

where n is the number of the data points, y_i is the observed value, $\acute{y}_i$ is the predicted value, $\bar{y}_i$ is the mean value of all observed data.

The value of R^2 close to 1 indicates that the fit is often a good one

[116]. However, a larger value of R^2 does not necessarily mean a better fit since the degrees of freedom can influence R^2. The value of R^2 will rise if more parameters are added to the models, but this does not imply a better fit [67]. In addition, R^2 is sensitive to extreme data points and also influenced by the range of the independent variable [117]. The use of R^2 is particularly inappropriate if the models are obtained by different transformations of the response scale [118]. Sole reliance on R^2 may fail to reveal important data characteristics and model inadequacies. Thus, R^2 is not an adequate criterion for evaluation of the fitting quality. The adjusted R^2 (Adj. R^2) overcomes these disadvantages and may be a better metric, which is expressed as [119].

$$\text{Adj}.R^2 = 1 - \left(1 - R^2\right)\left(\frac{n-1}{n-p}\right) \tag{56}$$

Obviously, Adj. R^2 overcomes the problem of the rise in R^2, especially for the dataset with a small sample size. Other frequently used error statistics are summarized in Table S2. However, it is not sufficient enough to evaluate the fitting quality by error statistics alone. The residual plot is a more reliable evaluation criterion than error statistics, which is recommended to further diagnose the fitting results [120]. It can be used to examine the underlying statistical assumptions (i.e. constant variance, independence of variables and normality of the distribution) about residuals and provide valuable information on how to improve the models. If the fitting quality is acceptable, all residuals should tend to fall in a horizontal band centered around zero, showing no systematic tendencies toward a clear pattern [28]. Statistically speaking, rather than asking whether a particular fitting result is good, it is more appropriate to compare two fitting results. Current studies lack further analysis of the fitting results. In this work, the *F*-test and Akaike's information criterion (AIC) are used to compare the fitting results from two models for the same dataset, which are expressed as [121,122].

$$F = \frac{(\mathrm{RSS}_1 - \mathrm{RSS}_2)/(\mathrm{df}_1 - \mathrm{df}_2)}{\mathrm{RSS}_2/\mathrm{df}_2} \tag{57}$$

$$\mathrm{AIC} = \begin{cases} \mathrm{n}\ln\left(\frac{\mathrm{RSS}}{\mathrm{n}}\right) + 2\mathrm{p} & \left(\frac{\mathrm{n}}{\mathrm{p}} \geq 40\right) \\ \mathrm{n}\ln\left(\frac{\mathrm{RSS}}{\mathrm{n}}\right) + 2\mathrm{p} + \frac{2\mathrm{p}(\mathrm{p}+1)}{\mathrm{n}-\mathrm{p}-1} & \left(\frac{\mathrm{n}}{\mathrm{p}} < 40\right) \end{cases} \tag{58}$$

where RRS denotes the residual sum of square, df is the degree of freedom, the subscripts 1 and 2 correspond to the simple and complex models, respectively.

The *F*-test and AIC are available to determine which model is the better and thereby improve the predictive and explanatory abilities of the model. The *F*-test assumes that two models are nested, where one model is a simplified version of the other [123]. It takes advantage of the difference in RSS of each fit to find out which model is the best. The *F*-test is commonly used in linear regression or analysis of variance, which can examine the significant differences between two nested models. The significance of *F*-test consists in determining whether the regression coefficients in the model are significantly different and whether the model itself is significant. The cumulative distribution function of the *F*-distribution is assessed to yield a *p* value. If the *p* value is statistically significant ($p < 0.05$), the fitting ability of the complex model is better than the simpler model [122]. Otherwise, the complex model is rejected. The *F*-test does not decide which model is correct and provides information on the goodness of fit [123]. The objective of AIC is to find out a model that can best interpret the experimental data but contain the fewest model parameters. AIC can provide robust and precise estimates based on maximum likelihood to rank the models rather than concept of significance [124]. AIC has a larger penalty for overfitting and thus helps to avoid selecting overly complex models. The significance of AIC is to obtain a compromise of the complexity of the estimated models and their goodness of fit. Compared with the *F*-test, AIC can compare nested or non-nested models. Hence, any two models can be compared by AIC and the one with a smaller value of AIC is suggested to be optimal. The Akaike's weight (W_A) indicates the probability of a better model and can be used to further confirm the optimal model, which can be expressed as [125].

$$W_A = \frac{1}{1 + exp(0.5\Delta\mathrm{AIC})} \tag{59}$$

where ΔAIC is the difference between the AIC values of one model and the best model.

Taking the Bohart–Adams and fractal-like Bohart–Adams models as an example, this work aims to compare the fitting results of ciprofloxacin [126] and phenol and *p*-nitrophenol [86] adsorption by using OriginPro software. The *F*-test and AIC can be obtained by the following procedures. Step 1: perform the curve fitting twice by the Bohart–Adams and fractal-like Bohart–Adams models for the same dataset and create two fitting reports automatically. Step 2: select **Analysis** → **Fitting** → **Compare Model** from the Origin menu, and then open **Fitting: fitcmpmodel** dialog box. Step 3: specify the way to recalculate and update the result, fitting results, comparison method, fit parameters, fit statistics, and click **OK** (see Fig. S1). The results of model comparison will show in the **Book** dialog box. As shown in Fig. 5, the breakthrough curve of ciprofloxacin adsorption shows an asymmetric distribution. Compared with the Bohart–Adams model, the fitted curve provided by the fractal-like Bohart–Adams model is more consistent with the experimental data. Its predicted values are very close to the observed values. The corresponding residuals fluctuate randomly in the vicinity of zero and fall in a narrow horizontal band. It is observed from Table 2 that the values of RSS, reduced χ^2 and RMSE are smaller and that R^2 and Adj. R^2 are larger for the fractal-like Bohart–Adams model. The *p* value for *F*-test is <0.05, indicating that this comparison is significant. The value of AIC is also smaller and its weight almost approaches unity for the fractal-like Bohart–Adams model. Consequently, ciprofloxacin adsorption follows the fractal-like Bohart–Adams model. By contrast, the breakthrough curves of phenol and *p*-nitrophenol adsorption show a symmetric distribution. Thus, the Bohart–Adams and fractal-like Bohart–Adams models have comparable fitting abilities and the fitted curves are almost coincident. The fractal-like Bohart–Adams model is superior to the Bohart–Adams model in terms of various error statistics for phenol adsorption. It is worth noting that all error values were fairly close for *p*-nitrophenol adsorption. In this case, error statistics are not sufficient enough to evaluate the fitting quality. The *F*-test and AIC are particularly important for further diagnosis of the fitting results. One can readily see that the *p* value for *F*-test is much more than 0.05. The Bohart–Adams model has smaller value of AIC and larger value of W_A. This case indicates that a model with more parameters does not necessarily mean a better fit.

5. Existing problems

5.1. Data quality

High quality of the experimental data is a prerequisite for obtaining a good fit. In many cases, the data quality is purposely or unconsciously neglected when the theoretical models are used to analyze the breakthrough curves. The poor data quality is not conducive to the accurate prediction of the model parameters and insights into the dynamic adsorption behaviors in a fixed-bed column. To our knowledge, the relevant situations mainly include that: (i) Many studies lack the indispensable repeated experiments (no error bars); (ii) The measured breakthrough curves are not complete (not S-shaped) [48]; (iii) The data points fluctuate greatly [127]; and (iv) The breakthrough curves intersect under different experimental conditions [128]. In our opinion, the first two cases can be easily solved by the standardized operation procedures. The reason why the latter two cases exist is that the hydration process (volume expansion) occurs when the adsorbent is in contact with the influent or the dominant mass-transfer and reaction mechanisms change during the adsorption process.

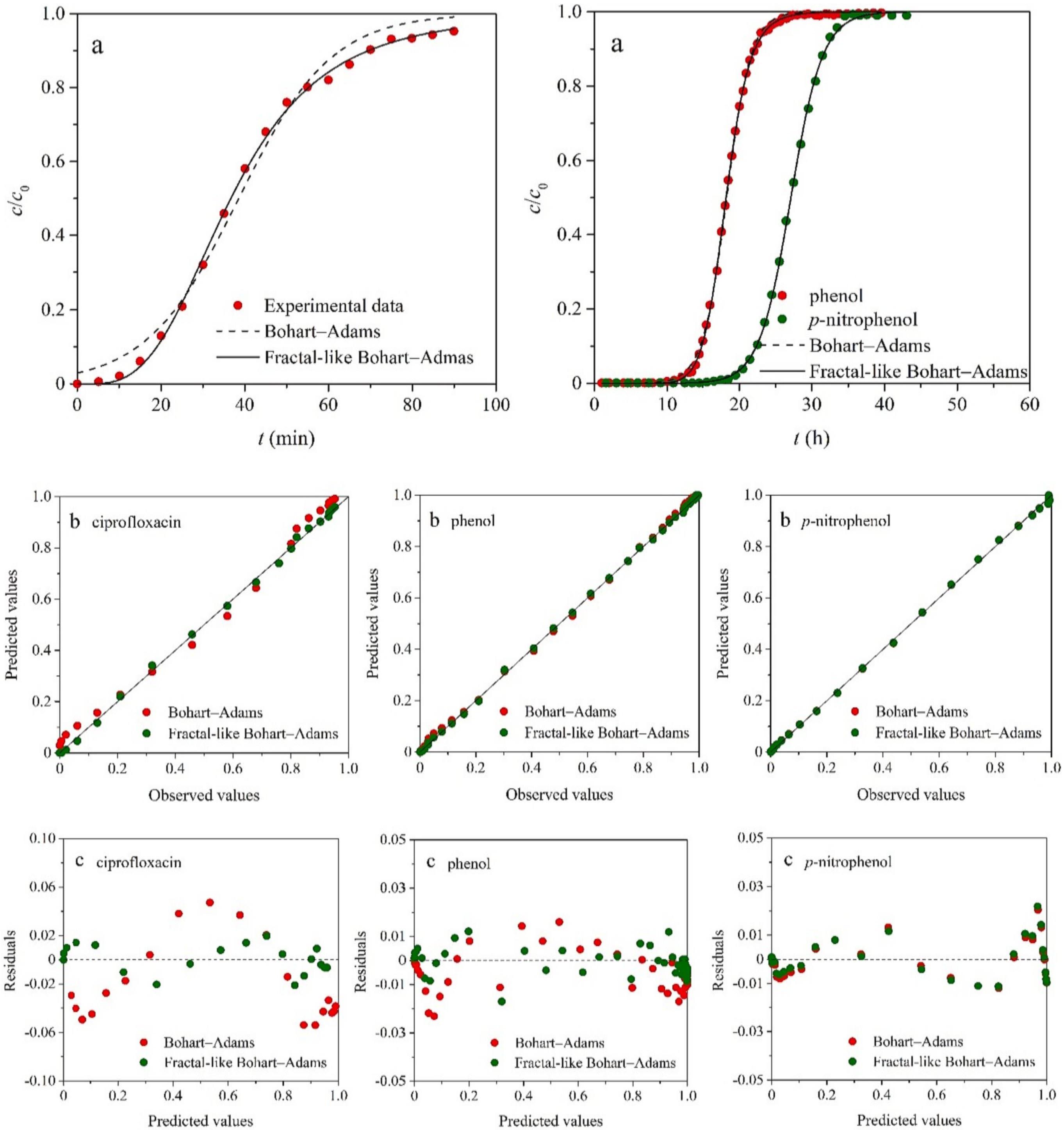

Fig. 5. Comparison of Bohart–Adams and fractal-like Bohart–Adams models: (a) fitted curves, (b) predicted versus observed values and (c) residual plot.

5.2. Partial and complete breakthrough curves

Since it is often a tedious and time-consuming work for obtaining the complete breakthrough curve, some studies focus more on how many bed volumes are required for breakthrough to occur [129]. Although the partial breakthrough curve can be obtained in a relatively short time, the completeness of the experimental data can directly influence the model parameters to be estimated. Besides, the partial breakthrough curve does not profoundly reveal the dynamic adsorption behaviors in a fixed-bed column. Taking the Thomas model (Eq. (14)) as an example, this work examines the fitting results of the partial and complete breakthrough curves from adsorption of methylene blue [37]. As shown in Fig. 6, the fitted curves provided by the Thomas model agree well with the partial and complete breakthrough curves at different influent concentrations and an error statistic Adj. R^2 is >0.99 in all cases. The fitting results of other partial breakthrough curves refer to Fig. S3. However, there are significant differences in k_T and q_0 obtained from the partial and complete breakthrough curves. As shown in Table 3, compared with the complete breakthrough curves, the maximum relative errors of k_T are 110.8 %, 26.1 % and 47.5 % respectively at 50, 100 and 150 mg L^{-1}, while the corresponding maximum relative errors of q_0 are −20.5 %, −4.0 % and −13.4 %. In general, the curve fitting aims to purely

Table 2
Fitting results of ciprofloxacin, phenol and *p*-nitrophenol adsorption with Bohart–Adams and fractal-like Bohart–Adams models.

Parameters and error statistics	Ciprofloxacin[a]		Parameters and error statistics	Phenol[b]		Parameters and error statistics	*p*-Nitrophenol[b]	
	Bohart–Adams	Fractal-like Bohart–Adams		Bohart–Adams	Fractal-like Bohart–Adams		Bohart–Adams	Fractal-like Bohart–Adams
k_{BA} (L mg min^{-1})	4.03×10^{-4}	–	k_{BA} (L mmol h^{-1})	0.113	–	k_{BA} (L mmol h^{-1})	8.64×10^{-2}	–
$k_{BA,0}$ (L mg min^{h-1})	–	1.17×10^{-3}	$k_{BA,0}$ (L mmol h^{h-1})	–	0.222	$k_{BA,0}$ (L mmol h^{h-1})	–	9.85×10^{-2}
a_0 (mg L^{-1})	669	1245	a_0 (mmol L^{-1})	594	1023	a_0 (mmol L^{-1})	882	936
h	–	0.487	h	–	0.422	h	–	5.82×10^{-2}
RRS	2.76×10^{-2}	2.47×10^{-3}	RRS	4.54×10^{-3}	1.52×10^{-3}	RRS	1.75×10^{-3}	1.77×10^{-3}
Reduced χ^2	1.62×10^{-3}	1.55×10^{-4}	Reduced χ^2	8.56×10^{-5}	2.92×10^{-5}	Reduced χ^2	5.16×10^{-5}	5.38×10^{-5}
R^2	0.9890	0.9990	R^2	0.9995	0.9998	R^2	0.9997	0.9997
Adj. R^2	0.9883	0.9989	Adj. R^2	0.9995	0.9998	Adj. R^2	0.9997	0.9997
RMSE	4.03×10^{-2}	1.24×10^{-2}	RMSE	9.25×10^{-3}	5.41×10^{-3}	RMSE	7.18×10^{-3}	7.33×10^{-3}
F-test	162.5 ($p = 8.55 \times 10^{-10} < 0.05$)		*F*-test	103.2 ($p = 6.03 \times 10^{-14} < 0.05$)		*F*-test	0.413 ($p = 0.525 > 0.05$)	
AIC	−116.6	−159.1	AIC	−510.7	−568.5	AIC	−350.7	−347.7
W_A	5.71×10^{-10}	1.000	W_A	2.80×10^{-13}	1.000	W_A	0.817	0.183

[a] operating conditions of $c_0 = 225$ mg L^{-1}, $u = 1.91$ mL min, $x = 25$ cm.
[b] operating conditions of $c_0 = 5.32$ mmol L^{-1}, $u = 34.9$ mL h^{-1}, $x = 5.71$ cm.

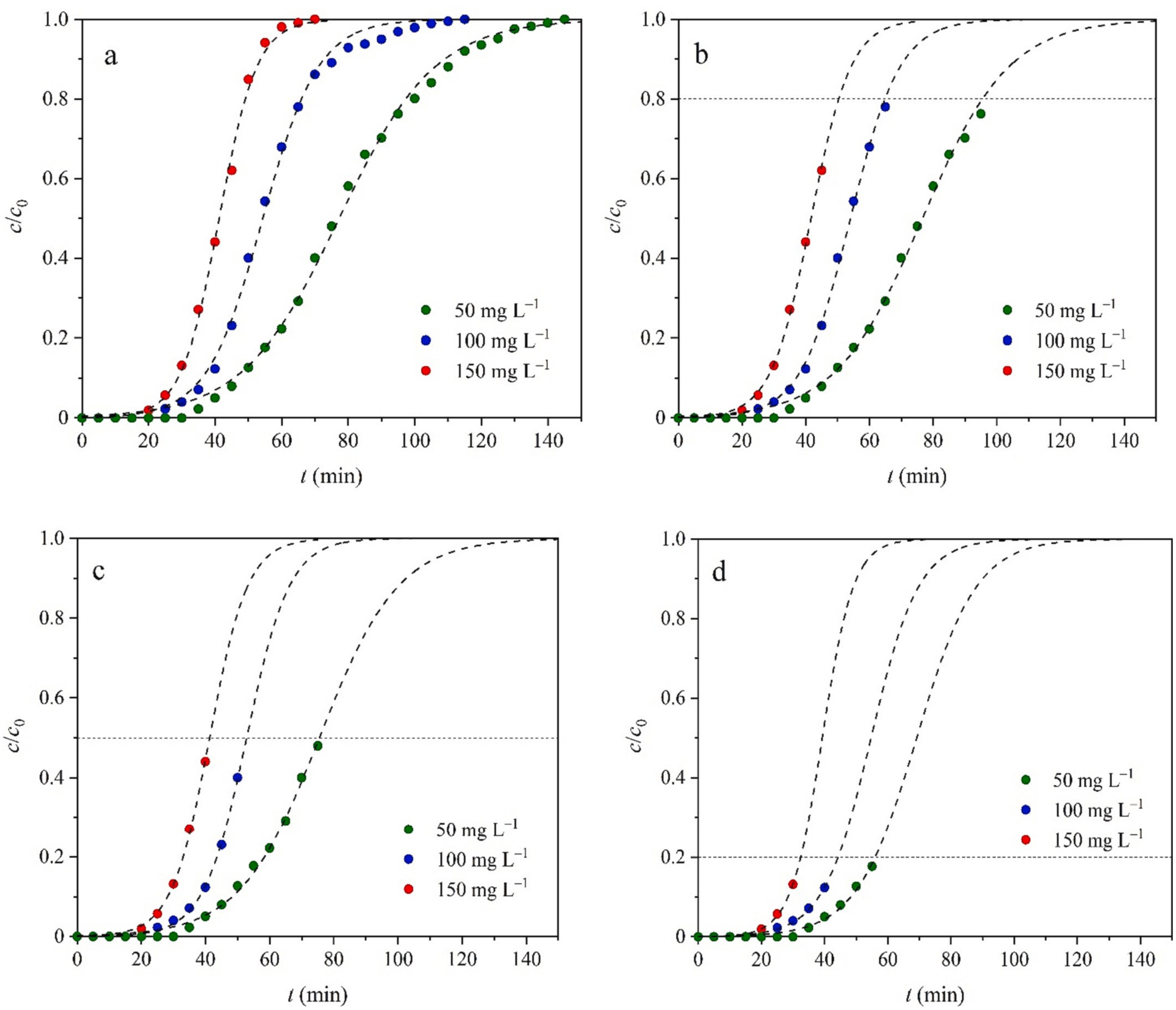

Fig. 6. Complete and partial breakthrough curves of methylene blue adsorption with the Thomas model: (a) 100 %, (b) 80 %, (c) 50 % and (d) 20 %. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Fitting results for complete and partial breakthrough curves of methylene blue adsorption at different concentrations.

c/c_0	50 mg L^{-1}			100 mg L^{-1}			150 mg L^{-1}		
	k_T (L mg^{-1} min^{-1})	q_0 (mg g^{-1})	Adj. R^2	k_T (L mg^{-1} min^{-1})	q_0 (mg g^{-1})	Adj. R^2	k_T (L mg^{-1} min^{-1})	q_0 (mg g^{-1})	Adj. R^2
1.0	1.39×10^{-3}	52.20	0.9977	1.19×10^{-3}	73.42	0.9978	1.18×10^{-3}	83.55	0.9984
0.9	1.40×10^{-3}	52.16	0.9957	1.23×10^{-3}	73.28	0.9975	1.14×10^{-3}	83.73	0.9966
0.8	1.48×10^{-3}	51.80	0.9959	1.29×10^{-3}	73.02	0.9977	1.06×10^{-3}	84.42	0.9980
0.7	1.57×10^{-3}	51.28	0.9970	1.33×10^{-3}	72.72	0.9978	1.06×10^{-3}	84.42	0.9980
0.6	1.60×10^{-3}	51.12	0.9961	1.40×10^{-3}	72.25	0.9978	1.12×10^{-3}	83.64	0.9974
0.5	1.62×10^{-3}	51.00	0.9941	1.50×10^{-3}	71.37	0.9979	1.12×10^{-3}	83.64	0.9974
0.4	1.67×10^{-3}	50.61	0.9914	1.50×10^{-3}	71.37	0.9979	1.24×10^{-3}	81.67	0.9978
0.3	1.66×10^{-3}	50.70	0.9842	1.40×10^{-3}	72.55	0.9958	1.24×10^{-3}	81.67	0.9978
0.2	2.11×10^{-3}	46.70	0.9826	1.33×10^{-3}	73.90	0.9872	1.35×10^{-3}	79.57	0.9933
0.1	2.93×10^{-3}	41.48	0.9654	1.48×10^{-3}	70.50	0.9681	1.74×10^{-3}	72.36	0.9962

minimize the deviations between the fitted curves and the data points when a mathematical model is used to analyze the breakthrough curves, ignoring the physical meanings of the model parameters. As a consequence, the model parameters obtained from the partial breakthrough curves are not applied to the design and optimization of the fixed-bed adsorber. On the whole, the complete breakthrough curves are recommended to obtain the model parameters.

5.3. Oversimplification of Bohart–Adams model

In addition to two forms of the Bohart–Adams model mentioned above, an oversimplified form of the Bohart–Adams model frequently appears in the recent literature, which is expressed as [130].

$$\frac{c}{c_0} = exp\left(k_{BA}c_0t - \frac{k_{BA}a_0x}{u}\right) \tag{60}$$

The simplification of Eq. (11) to Eq. (60) is completely unreasonable. From a mathematical point of view, Eq. (60) is an exponential function rather than a logistic function through oversimplification of the Bohart-Adams model, which covers up the functionality of the original Bohart-Adams model and greatly weakens the ability to describe the breakthrough curve [131]. By comparing Eq. (22), Eq. (24) and Eq. (60), it is found that the Wolborska and oversimplified Bohart–Adams models are equivalent in mathematical nature, i.e. $k_{BA} = \frac{\beta_a}{a_0}$ or $\frac{\beta_a\varepsilon}{\rho q_0}$. The fitted curve provided by Eq. (60) seriously deviates from the complete breakthrough curve, especially at larger values of *t*. For this reason, the fitting ability of Eq. (60) is also increasingly questioned. A poor fit is bound to result in the inaccurate estimation of the model parameters, which is detrimental to understanding the dynamic adsorption behaviors and optimizing the design of the adsorber [54]. In our opinion, the exponential term $exp\left[k_{BA}c_0\left(\frac{a_0x}{uc_0} - t\right)\right]$ in Eq. (11) contains the independent valuable *t*, whose value decreases with the increase in *t*. This exponential term will be much <1 when the value of *t* is sufficiently large. In this case, the unity term is not neglected in the denominator of Eq. (11). Hence, the use of Eq. (60) should be avoided in the future studies. In order to generalize the Bohart–Adams model and enable it to describe the asymmetric breakthrough curve, Hu et al. derive the *n*-order Bohart–Adams model, which is expressed as [104].

$$\frac{c}{c_0} = \left[1 + na_0^{1-n}c_0^{n-1}\left(\left[\frac{1+(n-1)\frac{k_na_0c_0^{n-1}x}{u}}{1+(n-1)k_na_0^{n-1}c_0t}\right]^{\frac{1}{n-1}} - \left[\frac{1}{1+(n-1)k_na_0^{n-1}c_0t}\right]^{\frac{1}{n-1}}\right)\right]^{-\frac{1}{n}} \tag{61}$$

5.4. Equivalent models

In some recent papers [132,133], the Bohart–Adams, Thomas and Yoon–Nelson models show different fitting results (fitted curves and error values) when used to analyze the breakthrough curves. In other words, these three models are assumed to be separate and independent. To our knowledge, A lack of the understanding of their intrinsic mathematical relationships can account for the further propagation of such a mistake. The previous study has demonstrated that these three models were mathematically a logistic function that represents a symmetric S-shaped curve and their parameters are interchangeable, which are expressed as [1].

$$k_{YN} = k_{BA}c_0 = k_Tc_0 \tag{62}$$

$$\tau = \frac{a_0x}{uc_0} = \frac{q_0m}{\nu c_0} \tag{63}$$

Hence, the Bohart–Adams, Thomas and Yoon–Nelson models are equivalent. Their fitted curves should be coincident and all error values are equal when the curve fitting is carried out for the same set of the experimental data. The above relationships reveal the absurd behaviors of treating these three models as independent models and comparing their respective fitting abilities based on various error statistics. According to Eq. (62), the parameters k_{BA} and k_T are regarded as the second-order reaction rate constants. Similar to the parameter τ, the physical meanings of the terms $\frac{a_0x}{uc_0}$ and $\frac{q_0m}{\nu c_0}$ represent the operating time required to reach 50 % breakthrough [64]. According to Eq. (62) and Eq. (63), the determination of k_{BA}, k_T, a_0 and q_0 avoids the repeated curve fitting. The revelation of intrinsic relationships between the Bohart–Adams, Thomas and Yoon–Nelson models will contribute to precisely obtaining the model parameters and better understanding the dynamic adsorption behaviors in a fixed-bed column. Just as the roles of k_{YN} and τ, the terms $k_{BA}c_0$ (k_Tc_0) and $\frac{a_0x}{uc_0}$ $\left(\frac{q_0m}{\nu c_0}\right)$ determine the degree of curvature of the breakthrough curve and its location at the *t*-axis, respectively. The values of k_{YN} and τ depend on the operating conditions and to some extent may be regarded as the lumped parameters that embed some physical processes and operating features. From a mathematical perspective, the same breakthrough curve is obtained regardless of the process parameters only if the value of the terms $\frac{a_0x}{uc_0}$ and $\frac{q_0m}{\nu c_0}$ keeps constant.

5.5. Linear and nonlinear curve fitting

The accurate calculation of the model parameters plays an important role in evaluation the separation efficiency of the contaminant and identification of its mass-transfer law [134]. The values of the model parameters are affected by the regression method. Although it is highly questioned, the linear fitting is still frequently adopted by many researchers. The linearization of a model requires the corresponding transformation of the measured experimental data. The linearized treatment implicitly alters their error structure and may also violate the error variance and normality assumptions of standard least squares [135]. As a consequence, different linearized forms of a model may result in different estimates of the undetermined parameters [136]. In addition, the linearized treatment may cause arbitrary exclusion of certain data points from the estimated model for calculation as these data points are not in the domain of definition of the linearized equations. For instance, a linearized form of the Thomas model is expressed as.

$$ln\left(\frac{c_0}{c}-1\right)=\frac{k_T q_0 m}{\nu}-k_T c_0 t \quad (64)$$

It is universally acknowledged that the domain of definition of the logarithmic function is always more than zero and the bottom number of the power functions is not equal to zero. According to Eq. (64), one can get $0 < c < c_0$ or $0 < c/c_0 < 1$. When the Thomas model is used to fit the experimental data, the linearized treatment excludes data points at $c = 0$ and $c = c_0$. A new dependent variable $ln\left(\frac{c_0}{c}-1\right)$ is produced after linearization and its values may not be sufficiently accurate during the calculation process using the measured data. The linearization of a breakthrough model also results in the statistical bias. The error increases dramatically when the value of the logarithmic term in Eq. (64) is very close to its limit, especially for the breakthrough curve before breakthrough point and after saturation point [131]. Besides, the imprecisions of slope and intercept obtained from Eq. (64) may also be incorporated into the model parameters during the curve fitting. Last but not least, the linear fitting fails to examine more complex models that may be in better agreement with the experimental data. The nonlinear fitting is a versatile way to obtain the model parameters without any transformation [137]. It not only overcomes the existing disadvantages of the linear fitting, but also can solve the models with multiple parameters. The nonlinear fitting can provide a more robust and reliable parameter estimation than the linear fitting.

5.6. Reasons for asymmetric breakthrough curve

The breakthrough behaviors in a fixed-bed column are subject to the operating parameters, geometric dimensions of the column, types of the contaminants and physicochemical properties of the adsorbent. It is reported that the complete breakthrough curve is usually asymmetric S-shaped even for the single-solute adsorption [105]. When the solution containing a solute with a high affinity is fed into the fixed-bed column at high concentrations, the breakthrough curve first shows a sharp rise and then a slow approach to saturation. Such early breakthrough and tailing are caused by the slow surface diffusion [138]. If the intraparticle diffusion controls the adsorption kinetics, as it is often the case in practice, the breakthrough curve is asymmetric with a long tailing [33]. Moreover, an asymmetric breakthrough curve may be also attributed to the fact that the adsorbent contains two or more constituents of unequal reactivity or the rate of adsorption falls off more rapidly than the residual capacity of the adsorbent [14]. Thus, the measured breakthrough curves are asymmetric in most cases, which account for a relatively poor fit for the Bohart–Adams, Thomas and Yoon–Nelson models.

6. Concluding remarks

The mathematical modeling of the breakthrough curves is a practically indispensable link for the design and optimization of a fixed-bed reactor. However, the traditional breakthrough models are not applicable for the diffusion-limited heterogeneous processes due to their implicit assumptions of time-independent rate constants and transport properties. The fractal-like theory provides new insights into the dynamic adsorption behaviors and allows to understand the mechanisms controlling the process evolution deeply on a microscopic scale. Furthermore, based on the similarity of the breakthrough curves and some S-shaped function curves, the empirical breakthrough models developed have been an alternative strategy for modeling of the measured breakthrough curves. In most cases, the complete breakthrough curves are asymmetric S-shaped even for the single-solute adsorption. During the curve fitting, high quality of the experimental data is the first step to obtaining a good fit. The selection of an appropriate mathematical model for prediction of the breakthrough curves requires not only taking into account its underlying assumptions, curve characteristics and application scope, but also allowing for data quality, fitting methods, error statistics and residual plot. *F*-test and AIC can be used to compare the fitting results of two breakthrough models for the same dataset, but the former requires the two models must be nested. The breakthrough models with asymmetric properties usually have higher fitting quality. The partial breakthrough curves result in the inaccurate estimation of the model parameters. The Bohart–Adams, Thomas and Yoon–Nelson models are mathematically equivalent and their parameters are interchangeable. The Wolborska and over-simplified Bohart–Adams models are mathematically an exponential function, which are not applicable for prediction of the complete breakthrough curves. The fitting quality of the Chern–Chien model can be evaluated by the residual plot rather than error statistics since it adopts the iteration algorithm of ODR. This review is expected to help readers better understand and use the breakthrough models with simple analytical solutions and avoid the further propagation of some existing mistakes and inconsistencies in the future studies.

CRediT authorship contribution statement

Qili Hu: Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Xingyue Yang:** Validation, Formal analysis, Data curation. **Leyi Huang:** Validation, Software, Data curation. **Yixi Li:** Validation, Software, Formal analysis. **Liting Hao:** Validation, Funding acquisition, Data curation. **Qiuming Pei:** Software, Formal analysis, Data curation. **Xiangjun Pei:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jwpe.2024.105065.

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